

Software Requirements Specification

for Online Exam Portal

Version 1.0 Draft

Prepared by Team 6: Chathurth R, Darshana S, Arati, Prakash N Heroor

PES University, RR Campus

7 September 2026

Table of Contents

1. Introduction.....	4
1.1 Purpose.....	4
1.2 Intended Audience and Reading Suggestions	4
1.3 Product Scope.....	4
1.4 References	4
2. Overall Description.....	4
2.1 Product Perspective	4
2.2 Product Functions	4
2.3 User Classes and Characteristics	5
2.4 Operating Environment.....	5
2.5 Design and Implementation Constraints	5
2.6 Assumptions and Dependencies	5
3. External Interface Requirements	5
3.1 User Interfaces	5
3.2 Software Interfaces	5
3.3 Communications Interfaces	5
4. Analysis Models	5
4.1 Use Case Diagram.....	6
4.2 Entity-Relationship Diagram	6
5. System Features.....	6
5.1 User Authentication	6
5.2 Exam Management	7
5.3 Exam Taking and Evaluation	8
5.4 Results	9
6. Other Nonfunctional Requirements.....	9
6.1 Performance Requirements.....	9
6.2 Safety Requirements	9
6.3 Security Requirements	10
6.4 Software Quality Attributes	10
6.5 Business Rules.....	10
7. Other Requirements.....	10
Appendix A: Glossary	11
Appendix B: Field Layouts	11
Appendix C: Requirement Traceability Matrix.....	11

Revision History

Name	Date	Reason For Changes	Version
Darshana S	7-09-2026	Initial draft (11 requirements: 7 functional, 4 non-functional)	1.0 Draft
Arati	30-09-2026	Updated Appendix C (Requirement Traceability Matrix) by replacing TBD entries with the corresponding Test Case IDs from the Test Plan.	1.1 Draft

1. Introduction

1.1 Purpose

This SRS specifies the requirements for the Online Exam Portal, version 1.0. It covers the complete web application: user accounts, MCQ and exam creation, timed exam delivery, automatic evaluation and result viewing.

1.2 Intended Audience and Reading Suggestions

- Developers (Arati, Prakash): Sections 3, 5 and 6.
- Project manager (Chathurth): Sections 1, 2 and 5.
- Testers and documentation (Darshana): Sections 5 and 6 and Appendix C. Test cases are derived from the requirement IDs (FR-1 to FR-7, NFR-1 to NFR-4).
- Instructors and faculty reviewers: Sections 1 and 2.

Section 2 gives the overall description, Section 3 the interfaces, Section 4 the analysis models, Section 5 the functional requirements, and Section 6 the non-functional requirements.

1.3 Product Scope

The Online Exam Portal is a web-based system where instructors create multiple-choice exams and students take them within a timed window. The system grades responses automatically and shows results. Objectives:

- Replace paper-based MCQ tests with an online process.
- Give instant, accurate scoring.
- Give instructors a simple view of student performance.

1.4 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications
- Course-provided SRS and Test Plan templates, PES University
- Django documentation, <https://docs.djangoproject.com>

2. Overall Description

2.1 Product Perspective

The Online Exam Portal is a new, self-contained web application. A Django application serves the pages and stores data in a SQLite database. Users access it through a web browser.

2.2 Product Functions

- Registration, login and role-based access (Student, Instructor)
- MCQ question management
- Exam creation and scheduling
- Timed exam attempt and submission
- Automatic evaluation and scoring
- Result viewing for students and class reports for instructors
- User management through the built-in Django admin site (not a separate requirement)

2.3 User Classes and Characteristics

User Class	Description	Importance
Student	Takes exams and views own results; limited technical expertise	High
Instructor	Creates questions and exams, views class results; moderate technical expertise	High
Administrator	Manages accounts through the Django admin site; created by the developers, not through registration	Low

2.4 Operating Environment

- Client: current versions of Chrome and Firefox on Windows, Linux or macOS.
- Server: Python 3.11+ and Django 5.x on Windows or Linux.
- Database: SQLite.

2.5 Design and Implementation Constraints

- Backend in Python (Django); pages served by the Django application using HTML, CSS and JavaScript. JavaScript is used for the countdown timer and question navigation.
- Authentication uses Django sessions.
- Only multiple-choice questions with one correct answer are supported in v1.0.
- Development follows the semester timeline and uses the team's public GitHub repository.

2.6 Assumptions and Dependencies

- Students have a stable internet connection during exams.
- The server clock is the authoritative time source for all timers.
- All students may see every published exam; per-student enrollment is out of scope for v1.0.

3. External Interface Requirements

3.1 User Interfaces

- Screens: Register, Login, Student Dashboard, Exam Instructions, Exam Attempt (question, options, navigator, countdown timer, Previous/Next/Submit), Result, Instructor Dashboard, Question Form, Exam Form, Class Report.
- Every screen has a header with the user name and a Logout button.
- Validation errors appear next to the field they belong to.

3.2 Software Interfaces

- Database: SQLite accessed through the Django ORM.
- Browser: standard HTML forms and JavaScript.

3.3 Communications Interfaces

- HTTP between browser and server (localhost during development; HTTPS if deployed).

4. Analysis Models

4.1 Use Case Diagram

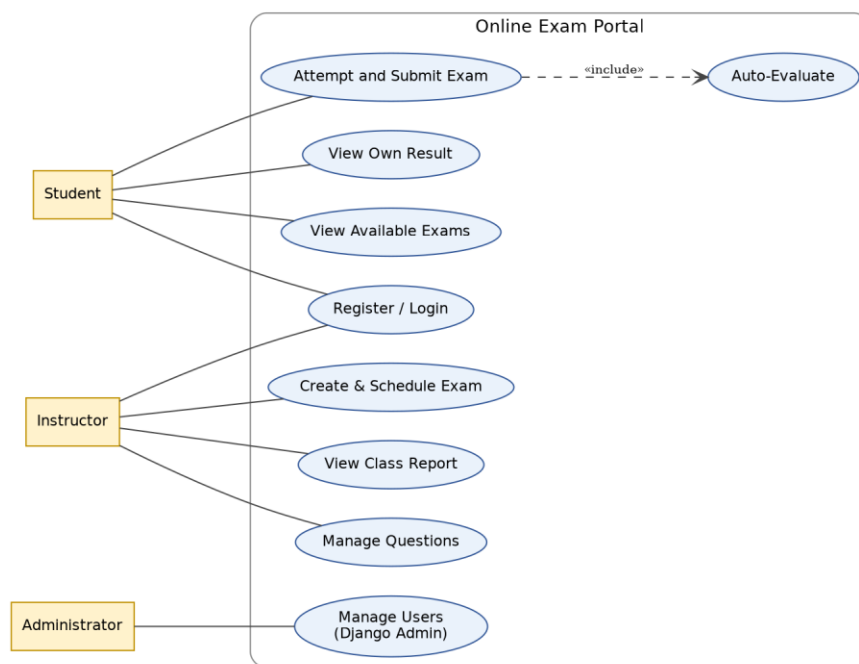


Figure 1: Use case diagram

4.2 Entity-Relationship Diagram

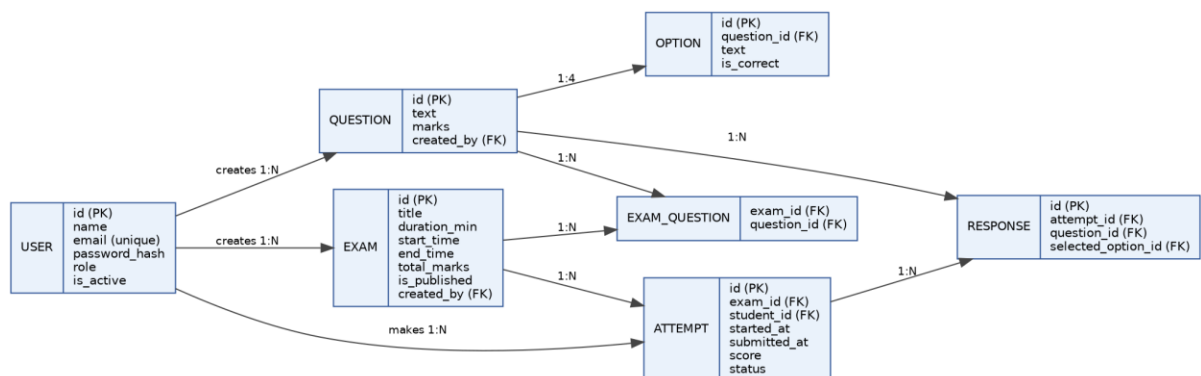


Figure 2: Entity-relationship diagram

5. System Features

5.1 User Authentication

5.1.1 Description and Priority

High priority. Users create accounts and log in; the system restricts pages by role.

5.1.2 Stimulus/Response Sequences

- User submits the registration form: system validates, creates the account and redirects to Login with a confirmation.
- User submits credentials: system authenticates and opens the dashboard for the role.
- Invalid credentials: system shows "Invalid email or password".

5.1.3 Functional Requirements

FR-1: User Registration

The system shall allow a new user to register as a Student or an Instructor.

- The form contains full name, email, password, confirm password and role; all fields are mandatory.
- The email must be in a valid format and not already registered.
- The password must have at least 8 characters with at least one letter and one digit.
- Password and confirm password must match.
- Each invalid field shows a message next to it, and no account is created.
- On success the account is created, the password is stored hashed, and the user is redirected to Login with a confirmation message.
- Administrator accounts cannot be created through registration.

FR-2: Login, Logout and Role-Based Access

The system shall authenticate users and restrict pages according to their role.

- Users log in with email and password and are redirected to the Student or Instructor dashboard.
- Wrong credentials show "Invalid email or password" without saying which one was wrong.
- Logout ends the session; the browser Back button then cannot open protected pages.
- Unauthenticated users opening a protected page are redirected to Login.
- A Student opening an instructor page, or an Instructor opening a student page, sees an access-denied (403) message.
- Deactivated accounts cannot log in.
- A session expires after 30 minutes of inactivity.

5.2 Exam Management

5.2.1 Description and Priority

High priority. Instructors build a bank of MCQs and use it to create scheduled exams.

5.2.2 Stimulus/Response Sequences

- Instructor submits a question form: system validates and saves it.
- Instructor submits an exam form with selected questions: system saves it as a draft.
- Instructor publishes the exam: it becomes visible to students.

5.2.3 Functional Requirements

FR-3: Create and Manage MCQ Questions

The system shall allow an Instructor to create, edit, delete and list their own multiple-choice questions.

- A question has text, exactly 4 options, exactly 1 correct option, and marks (a positive integer, default 1).
- The system rejects empty question text, any blank option, duplicate options, and a question with no correct option.
- An Instructor sees, edits and deletes only their own questions.
- A question used in an exam that already has attempts cannot be edited or deleted.

- Deleting a question shows a confirmation and removes it from draft exams.

FR-4: Create and Schedule an Exam

The system shall allow an Instructor to create an exam from their questions, set its schedule and publish it.

- An exam has a title, duration in minutes (1 to 180), start date-time, end date-time, and at least one selected question.
- The end time must be after the start time, and the start time cannot be in the past when creating.
- The duration cannot exceed the length of the window between start and end time.
- Total marks are calculated automatically as the sum of the selected questions' marks.
- An exam stays a draft until published; only published exams are visible to students.
- An exam with attempts cannot be edited or deleted.
- An Instructor sees, edits and deletes only their own exams.

5.3 Exam Taking and Evaluation

5.3.1 Description and Priority

High priority. Students take a published exam inside its window; the system grades the attempt automatically.

5.3.2 Stimulus/Response Sequences

- Student opens the dashboard: system lists published exams with their status.
- Student starts an open exam: system creates the attempt and starts the timer.
- Student submits, or the timer reaches zero: system locks the attempt, scores it and shows the result.

5.3.3 Functional Requirements

FR-5: Attempt an Exam

The system shall let a Student view available exams and attempt an open exam under a timer.

- The Student dashboard lists published exams with title, start, end, duration and status (Upcoming, Open, Closed, Completed).
- A Student can start an exam only while it is Open, that is between its start and end time.
- Each Student can attempt an exam only once; a second start attempt is refused with a message.
- An instructions page shows duration, number of questions and total marks before the exam starts.
- The attempt page shows one question at a time with 4 options, Previous/Next buttons and a question navigator marking answered questions.
- A countdown timer runs from the server-side start time; the attempt ends at the duration or at the end of the exam window, whichever is earlier.
- A Student can change answers before submitting and can leave questions unanswered.

FR-6: Submit an Exam and Automatic Evaluation

The system shall accept the submission of an attempt and evaluate it automatically.

- Manual submit asks for confirmation and shows the number of unanswered questions.

- When the timer reaches zero the attempt is submitted automatically with the answers selected so far.
- After submission the attempt is locked; Back or reload cannot change answers.
- The server rejects any answer received after the deadline.
- Each correct answer earns the question's full marks; wrong and unanswered questions earn zero (no negative marking).
- The total score and percentage are stored and calculated exactly once, at submission.
- The result page opens immediately after submission.

5.4 Results

5.4.1 Description and Priority

Medium priority. Students see their own results; Instructors see how a class performed.

5.4.2 Stimulus/Response Sequences

- Student opens My Results: system lists their attempts with scores.
- Instructor opens an exam report: system lists all attempts with summary figures.

5.4.3 Functional Requirements

FR-7: View Results

The system shall show a Student their own results and an Instructor the results of their own exams.

- A Student sees score, total marks, percentage and a Pass/Fail status (pass mark 40%) for each submitted attempt.
- A Student can list all their past attempts with dates and scores.
- A per-question review (own answer against correct answer) becomes available only after the exam window closes.
- No result is shown for an exam that has not been submitted.
- An Instructor sees a report per exam listing each student's name, email, score, percentage and submission time.
- The report shows the number of attempts and the average, highest and lowest score.
- A Student cannot see other students' results, and an Instructor cannot see reports of other instructors' exams.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

NFR-1: Performance

The system shall respond quickly on the demonstration setup.

- Normal pages load within 3 seconds on the demonstration machine.
- Evaluation finishes and the result page opens within 2 seconds of submission for exams of up to 50 questions.
- The system supports at least 10 students taking an exam at the same time.

6.2 Safety Requirements

NFR-2: Safety (Data Protection)

The system shall protect exam data from loss or unintended change.

- A submitted attempt is saved to the database before its result is shown, so a later server restart does not lose it.
- Submitted attempts and scores cannot be edited or deleted by Students or Instructors through the application.
- A question or exam that has attempts cannot be deleted, so past results are never orphaned.
- If saving a submission fails, the system shows an error and keeps the attempt open so the Student can retry before the deadline.

6.3 Security Requirements

NFR-3: Security

The system shall protect accounts and exam data.

- Passwords are stored only as salted hashes (Django default hashing).
- Role checks are enforced on the server for every request, not only by hiding menu items.
- All forms are protected against CSRF, and database access uses the Django ORM (no raw SQL with user input).
- Correct answers are never sent to the browser while an exam is in progress.
- Protected pages require a valid session.

6.4 Software Quality Attributes

NFR-4: Usability and Compatibility

The system shall be easy to use and work in common browsers.

- Works in current versions of Chrome and Firefox with no plugins.
- Layouts remain usable from 360 px screen width upward.
- A Student can open and start an available exam within 3 clicks of logging in.
- Error and confirmation messages use a consistent style and wording on every screen.

6.5 Business Rules

- Only Instructors create questions and exams; only Students attempt exams.
- Each Student has exactly one attempt per exam.
- The pass mark is 40% of the total marks.
- Instructors access only their own questions, exams and reports.

Domain requirements: Exam timing always uses the server clock. Question marks are positive integers. An attempt is either In Progress or Submitted.

7. Other Requirements

- Database migrations are kept under version control in the repository.
- The SQLite database file is backed up before each demonstration (a team practice, not a tested software requirement).
- The code and all project documents live in the team's public GitHub repository.

Appendix A: Glossary

Term	Meaning
MCQ	Multiple-Choice Question
Attempt	One student's single sitting of an exam
Exam window	Time between the exam start and end date-times in which it can be started
FR / NFR	Functional / Non-functional Requirement
CSRF	Cross-Site Request Forgery

Appendix B: Field Layouts

Entity	Fields
User	id, name, email (unique), password_hash, role, is_active
Question	id, text, marks, created_by
Option	id, question_id, text, is_correct
Exam	id, title, duration_min, start_time, end_time, total_marks, is_published, created_by
ExamQuestion	exam_id, question_id
Attempt	id, exam_id, student_id, started_at, submitted_at, score, status
Response	id, attempt_id, question_id, selected_option_id

Appendix C: Requirement Traceability Matrix

Req ID	Requirement	Section	Priority	Test Case IDs
FR-1	User Registration	5.1	High	UT_01, UT_02, UT_03, UT_04, UT_05, UT_06, UT_07, UT_08, IT_01, ST_01
FR-2	Login, Logout and Role-Based Access	5.1	High	UT_09, UT_10, UT_11, UT_12, UT_13, IT_02, IT_03, ST_02, ST_03, ST_04
FR-3	Create and Manage MCQ Questions	5.2	High	UT_14, UT_15, UT_16, UT_17, UT_18, UT_19, UT_20, IT_04, IT_05, ST_05
FR-4	Create and Schedule an Exam	5.2	High	UT_21, UT_22, UT_23, UT_24, UT_25, UT_26, IT_06, IT_07, ST_06
FR-5	Attempt an Exam	5.3	High	UT_27, UT_28, UT_29, UT_30, ST_07, IT_08, UT_31, UT_32, ST_08, IT_09
FR-6	Submit an Exam and Automatic Evaluation	5.3	High	UT_33, IT_10, ST_09, ST_10, UT_34, UT_35, IT_11, ST_11
FR-7	View Results	5.4	Medium	UT_36, UT_37, ST_12, UT_38, IT_12, UT_39, ST_13, ST_14

NFR-1	Performance	6.1	Medium	PT_01, PT_02, PT_03
NFR-2	Safety (Data Protection)	6.2	Medium	ST_15, ST_16, ST_17
NFR-3	Security	6.3	High	ST_18, ST_19, ST_20, ST_21
NFR-4	Usability and Compatibility	6.4	Medium	UT_40, UT_41, UT_42